Tensor product miscellany

Soham Saha
https://soham-saha.github.io
https://soham-saha.codeberg.page

We will be working over $\mathbf{Vect}_k$ for arbitrary field k.

Define $T^n(V)$ as $\underbrace{V \otimes ... \otimes V}_{n}$.

$T^0(V)$ is defined as k, the scalar field interpreted as a vector space over itself.

If V has basis $\{u_1, ..., u_d\}$, then $T^n(V)$ has basis $\left\{u_{i_1} \otimes ... \otimes u_{i_n} : 1 \leq i_1, ..., i_n \leq d\right\}$.

Show that $T^n(V^*)$ is canonically isomorphic to $(T^n(V))^*$, when V is a finite dimensional vector space.

Given $f_1, ..., f_n \in V^*$, we can define a multilinear map from $(V, ..., V)$ to k as $(v_1, ..., v_n) \mapsto \prod f_i(v_i)$. We can compress the data of this multilinear map to a linear map from $T^n(V)$ to k. Therefore, for every $(f_1, ..., f_n) \in (V^*, ..., V^*)$, we have an object in $(T^n(V))^*$. Furthermore, this is a multilinear map from $(V^*, ..., V^*)$ to $(T^n(V))^*$. Finally, we can compress the data down to a linear map from $T^n(V^*)$ to $(T^n(V))^*$, which when written simply, is just

$$(f_1 \otimes ... \otimes f_n)(v_1 \otimes ... \otimes v_n) = f_1(v_1) \times ... \times f_n(v_n)$$

This map is injective, and since V is given to be finite dimensional, so by comparing dimensions we get that $(T^n(V))^*$ and $T^n(V^*)$ are isomorphic.

If V and W are infinite dimensional, then there can be injective maps from V to W which are not surjective. Therefore, the above doesn't work when V is infinite dimensional.

Wait, what about the map from $(T^n(V))^*$ to $T^n(V^*)$?

I am yet to express this "canonically", in the sense that I haven't yet found a definition which doesn't rely on a basis for V. But, it is still canonical, formally speaking, because the choice of the basis doesn't change the function that we get; i.e., whatever basis you choose for V in the following definition, the map $\varphi : T^n(V)^* \to T^n(V^*)$ will be always same.

Choose a basis $\{e_i\}_{1 \leq i \leq d}$ for V, and take the dual basis $\{e^i\}_{1 \leq i \leq d}$ for V^*. The map $\varphi : T^n(V)^* \to T^n(V^*)$ is given by

$$f \mapsto \sum_{i_1, ..., i_n = 1}^{d} f\left(e_{i_1} \otimes ... \otimes e_{i_n}\right) \cdot e^{i_1} \otimes ... \otimes e^{i_n}$$

The function φ stays same for every choice of basis $\{e_i\}$.

Note that if V is infinite dimensional, then $T^n(V^*) \ncong (T^n(V))^*$.

Not all $w \in T^n(V)$ are simple.

For example, $e_1 \otimes e_2 + e_2 \otimes e_1 \in T^2(\mathbb{R}^2)$ is non-simple.

Show how to give a ring structure over

$$T(V) \coloneqq \bigoplus_{i \geq 0} T^i(V)$$

Given $u_1 \otimes ... \otimes u_m \in T^m(V)$ and $v_1 \otimes ... \otimes v_n \in T^n(V)$, we define the product in the ring to be

$$(u_1 \otimes ... \otimes u_m) \times (v_1 \otimes ... \otimes v_n) = u_1 \otimes ... \otimes u_m \otimes v_1 \otimes ... \otimes v_n \in T^{m+n}(V)$$

T^n is functorial, k-linear maps from V to W extend easily to k-linear maps from $T^n(V)$ to $T^n(W)$.

Given a k-linear map $\varphi \in \mathrm{Hom}_k(V, W)$, we have a multilinear map from $(V, ..., V)$ to $T^n(W)$ given by $(\varphi(v_1), ..., \varphi(v_n)) \mapsto (v_1 \otimes ... \otimes v_n)$ (this is produced by composing the list $[\varphi, ..., \varphi]$ with the multimorphism from $(W, ..., W)$ to $T^n(W)$). This map then gives us a linear map from $T^n(V)$ to $T^n(W)$.

Note that T^0 is also functorial, with object map $A \mapsto (T^0 A = k)$ and morphism map $f \in \mathrm{Hom}(A, B) \mapsto (T^0 f = \mathrm{id}_k) \in \mathrm{Hom}(k, k)$.
